Customer Shopping Behavior Analysis

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1. Project Overview

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

2. Dataset Summary

- Rows: 3,900
- Columns: 18
- Key Features:
  - Customer demographics (Age, Gender, Location, Subscription Status)
  - Purchase details (Item Purchased, Category, Purchase Amount, Season, Size, Color)
  - Shopping behavior (Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type)
- Missing Data: 37 values in Review Rating column

3. Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discou Appli
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	39
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	22
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	N
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	N
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	N
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	N
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	N
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	N
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	N

Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
3900	3900	3900.000000	3900	3900
2	2	NaN	6	7
No	No	NaN	PayPal	Every 3 Months
2223	2223	NaN	677	584
NaN	NaN	25.351538	NaN	NaN
NaN	NaN	14.447125	NaN	NaN
NaN	NaN	1.000000	NaN	NaN
NaN	NaN	13.000000	NaN	NaN
NaN	NaN	25.000000	NaN	NaN
NaN	NaN	38.000000	NaN	NaN
NaN	NaN	50.000000	NaN	NaN

- **Missing Data Handling:** Checked for null values and imputed missing values in the `Review Rating` column using the median rating of each product category.
- **Column Standardization:** Renamed columns to **snake case** for better readability and documentation.
- **Feature Engineering:**
  - Created **age_group** column by binning customer ages.

```
# create a column age_group
labels = ['Young Adult','Adult','Middle aged','Senior']
df['age_group'] = pd.qcut(df['age'],q=4,labels = labels)

df[['age','age_group']].head(10)

   age    age_group
0   55  Middle aged
1   19  Young Adult
2   50  Middle aged
3   21  Young Adult
4   45  Middle aged
5   46  Middle aged
6   63       Senior
7   27  Young Adult
8   26  Young Adult
9   57  Middle aged
```

- Created **purchase_frequency_days** column from purchase data.

```
frequency_mapping = {'Fortnightly': 14,
                     'Weekly' : 7,
                     'Monthly' : 30,
                     'Quarterly' : 90,
                     'Bi_Weekly' : 14,
                     'Annually' : 365,
                     'Every 3 Months': 90}
df['purchase_frequency_days'] =
df['frequency_of_purchases'].map(frequency_mapping)


df[['purchase_frequency_days','frequency_of_purchases']].head(10)
```

```
   purchase_frequency_days frequency_of_purchases
0                     14.0            Fortnightly
1                     14.0            Fortnightly
2                      7.0                 Weekly
3                      7.0                 Weekly
4                    365.0               Annually
5                      7.0                 Weekly
6                     90.0              Quarterly
7                      7.0                 Weekly
8                    365.0               Annually
9                     90.0              Quarterly

df[['discount_applied','promo_code_used']].head(10)
```

- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to MySQL and loaded the cleaned DataFrame into the database for SQL analysis
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```
engine = create_engine(
    "mysql+pymysql://root:852852@127.0.0.1:3306/customer"
)


engine.connect()
print("MySQL Connected Successfully ")
df.to_sql(
    name="customer_behaviour_set",
    con=engine,
    if_exists="replace",

    index=False
)



MySQL Connected Successfully 

3900
```

.

4. Data Analysis using SQL (Business Transactions)

We performed structured analysis in PostgreSQL to answer key business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

```sql
-- What is the total revenue generate by male vs female
-- customer

select gender,sum(purchase_amount) as revenue
from customer.customer_behaviour_set
group by gender;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

gender	revenue
Male	157890
Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

```sql
-- Which customer used a discount but still spent more than
-- the average purchase amount

select customer_id, purchase_amount
from customer.customer_behaviour_set
where discount_applied = 'Yes' and purchase_amount >=
(select avg(purchase_amount) from customer.customer_behaviour_set);
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

4. **Shipping Type Comparison** – Compared average purchase amounts between Standard and Express shipping.

```sql
-- compare the average purchase amount between standards and express
-- shipping?

select shipping_type, avg(purchase_amount)
 from customer.customer_behaviour_set
 where shipping_type in('Standard','express')
 group by shipping_type;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

shipping_type	avg(purchase_amount)
Express	60.4752
Standard	58.4602

5. **Subscribers vs. Non-Subscribers** – Compared average spend and total revenue across subscription status.

```
-- Do subscribed customers spend more ? compare average spend and
-- total revenue between subscribers and non subscribers ?

select subscription_status,
count(customer_id) as Total_customers,
round(avg(purchase_amount),2) as avg_spend,
round(sum(purchase_amount),2) as Total_revenue
from customer.customer_behaviour_set
group by subscription_status
order by total_revenue,avg_spend;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

subscription_status	Total_customers	avg_spend	Total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

6. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

```
-- Which 5 products have the highest percentage of purchase with
-- discount applied ?

select item_purchased,
round(sum(case when discount_applied = 'Yes' Then 1 else 0 end) /
 count(*) *100,2) as discount_rate
 from customer.customer_behaviour_set
 group by item_purchased
 order by discount_rate desc limit 5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

```sql
-- segment customers in to new, returning and loyal based on their
-- total number of previous purchases, and show the count of each
-- segment ?

with customer_type as (select customer_id, previous_purchases,
case
    when previous_purchases = 1 then 'New'
    when previous_purchases between 2 and 10 then 'Returning'
    else 'Loyal'
    End as Customer_segment
from customer.customer_behaviour_set)

select customer_segment, count(*) as "Number of Customers"
from customer_type
group by customer_segment
```

Result Grid | Filter Rows:

Customer_segment	Number of Customers
Loyal	3116
Returning	701
New	83

8. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

9. **Top 3 Products per Category** – Listed the most purchased products within each category.

```sql
-- What are the top 3 most purchased products within each category ?

WITH item_counts AS (
    SELECT
        category,
        item_purchased,
        COUNT(customer_id) AS total_orders
    FROM customer.customer_behaviour_set
    GROUP BY category, item_purchased),
ranked_items AS (
    SELECT
        category,
        item_purchased,
        total_orders,
        ROW_NUMBER() OVER (
            PARTITION BY category
            ORDER BY total_orders DESC) AS item_rank
    FROM item_counts)
SELECT item_rank, category, item_purchased, total_orders
FROM ranked_items
WHERE item_rank <= 3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Co

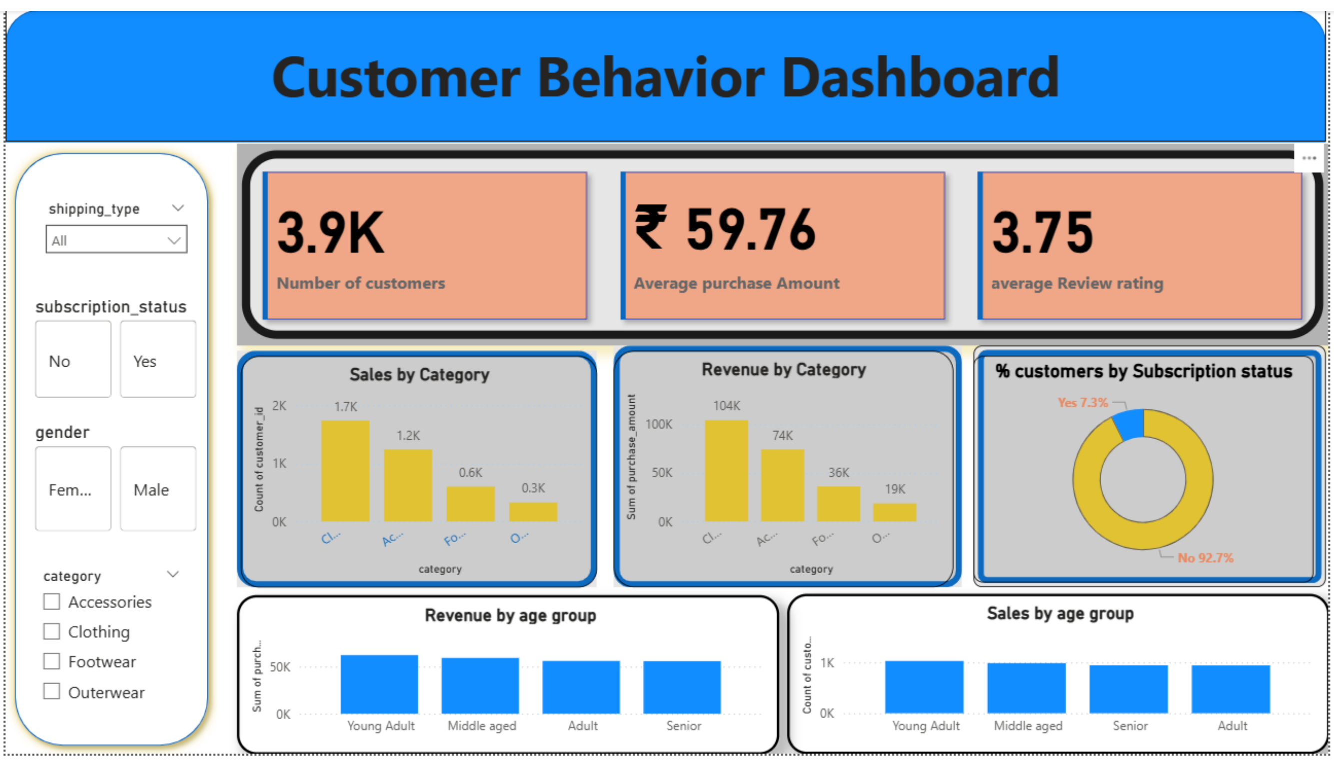
item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145

10. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

	age_group text	total_revenue numeric
1	Young Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

5. Dashboard in Power BI

Finally, we built an interactive dashboard in **Power BI** to present insights visually.



6. Business Recommendations

- **Boost Subscriptions** – Promote exclusive benefits for subscribers.

Key Business Insights

SUMMERY

- In this project, I analysed 3,900 customer records to identify purchasing patterns, customer loyalty trends, and revenue drivers
- I discovered that Clothing is the highest-performing category, Free Shipping significantly influences buying decisions, and a large portion of revenue comes from repeat customers.
- By segmenting customers based on age and purchase frequency, I identified opportunities for subscription growth and targeted marketing campaigns.
- The insights generated can help increase average order value, improve inventory planning, and enhance customer retention strategies.